

The following parameters are estimated :

- Armature resistance, R_a
- Back-emf constant, K_b
- Viscous friction coefficient, B
- Inertia, J
- Inductance, L

In addition, the same formulation also estimates the opposing external torque (system under study is a motor-generator set, so there exists a load torque because of the mechanical aspects), denoted by T_{gen} , which is assumed to be constant over the operating range considered.

1. Open Loop Test

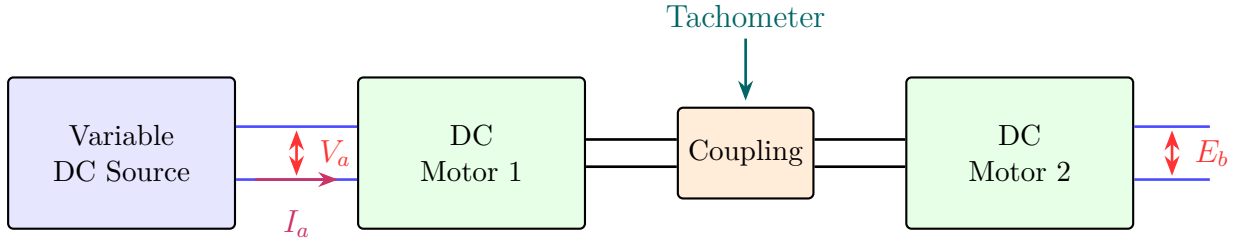


Figure 1: Block diagram of a open loop test

In this study, the motor is supplied from a variable DC source and operated at different voltage levels, ranging from zero up to its rated value. At each operating point, CSV data for the applied armature voltage (V_a), armature current (I_a), and back EMF (E_b) are recorded. The speed is measured using a tachometer for each operating point. **Note that all these measurements are taken at steady state condition.** Refer Figure 1

At steady state, the inductive term is neglected because:

$$L_a \frac{dI_a}{dt} \approx 0 \quad (1)$$

Thus, for each operating point:

$$\bar{V}_a = \bar{I}_a R_a + \bar{E}_b \quad (2)$$

1.1. Estimation of Armature Resistance R_a

From (2)

$$R_a = \frac{\bar{V}_a - \bar{E}_b}{\bar{I}_a} \quad (3)$$

Say we have taken N sets of measurement, then for each operating point "k" we can compute

$$R_{a,k}^{(\text{mean})} = \frac{\bar{V}_{a,k} - \bar{E}_{b,k}}{\bar{I}_{a,k}} \quad (4)$$

The final resistance used for subsequent computation is taken as the arithmetic mean of the each estimates:

$$R_a^{(\text{used})} = \frac{1}{N} \sum_{k=1}^N R_{a,k}^{(\text{mean})} \quad (5)$$

1.2. Estimation of Back-EMF Constant K_b

The back-emf relation is:

$$E_b = K_b \omega_m \quad (6)$$

Thus, for the k -th operating point:

$$K_b = \frac{\bar{E}_{b,k}}{\omega_{m,k}} \quad (7)$$

To find the final estimate, one can make use of the least square estimate through origin approach. Since the physical model implies zero back emf at zero speed, the best estimate is obtained by fitting a straight line through the origin:

$$\bar{E}_{b,k} = K_b \omega_{m,k} \quad (8)$$

Using least squares, the optimal slope is:

$$K_b = \frac{\sum_{k=1}^N \omega_{m,k} \bar{E}_{b,k}}{\sum_{k=1}^N \omega_{m,k}^2} \quad (9)$$

1.3. Estimation of Viscous Friction Coefficient B_m and Load Torque T_{gen}

We know

$$K_b I_a = B \omega_m + T_{gen} \quad (10)$$

Substituting ω_m from the electrical equation:

$$K_b I_a = B \left(\frac{V_a - I_a R_a^{(\text{used})}}{K_b} \right) + T_{\text{gen}} \quad (11)$$

Rearranging:

$$B = \frac{K_b^2 I_a}{V_a - I_a R_a^{(\text{used})}} - \frac{K_b}{V_a - I_a R_a^{(\text{used})}} T_{\text{gen}} \quad (12)$$

For the k -th operating point, define:

$$d_k = \frac{K_b^2 \bar{I}_{a,k}}{\bar{V}_{a,k} - \bar{I}_{a,k} R_a^{(\text{used})}} \quad (13)$$

$$\gamma_k = \frac{K_b}{\bar{V}_{a,k} - \bar{I}_{a,k} R_a^{(\text{used})}} \quad (14)$$

Then the friction coefficient satisfies:

$$B + \gamma_k T_{\text{gen}} = d_k \quad (15)$$

For all N valid operating points:

$$B + \gamma_1 T_{\text{gen}} = d_1$$

$$B + \gamma_2 T_{\text{gen}} = d_2$$

$$\vdots$$

$$B + \gamma_N T_{\text{gen}} = d_N$$

In matrix form:

$$\underbrace{\begin{bmatrix} 1 & \gamma_1 \\ 1 & \gamma_2 \\ \vdots & \vdots \\ 1 & \gamma_N \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} B \\ T_{\text{gen}} \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{bmatrix}}_{\mathbf{d}} \quad (16)$$

The unknown vector is:

$$\mathbf{x} = \begin{bmatrix} B \\ T_{\text{gen}} \end{bmatrix} \quad (17)$$

Using linear regression (least square approach), the above equations can be solved for the two unknowns.

1.4. Experimental Observations

Table 1: Steady-state PMDC parameter estimation data for each operating point

SL	$V_{a,\text{mean}}$ (V)	$I_{a,\text{mean}}$ (A)	$E_{b,\text{mean}}$ (V)	ω_m (rad/s)	$R_{\text{from_mean}}$ (Ω)	$K_{b,\text{point}}$ (V/(rad/s))	β_{point} (N·m·s/rad)
1	0.9592	0.9611	0.3541	13.6555	0.6297	0.02593	3.53×10^{-5}
2	1.9366	0.9998	1.3008	47.7836	0.6359	0.02722	3.21×10^{-5}
3	2.9250	1.0492	2.2420	82.0375	0.6510	0.02733	3.50×10^{-5}
4	3.9081	1.1118	3.1785	116.7625	0.6562	0.02722	3.92×10^{-5}
5	4.8989	1.1695	4.1029	151.6342	0.6807	0.02706	4.05×10^{-5}
6	5.8809	1.2201	5.0625	187.2389	0.6708	0.02704	4.01×10^{-5}
7	6.8572	1.2753	6.0523	223.0531	0.6311	0.02713	4.04×10^{-5}
8	7.8711	1.3021	7.0302	258.8672	0.6459	0.02716	3.76×10^{-5}
9	8.8476	1.3459	7.9874	294.0531	0.6391	0.02716	3.71×10^{-5}
10	9.8837	1.4065	8.9427	329.9719	0.6690	0.02710	3.81×10^{-5}

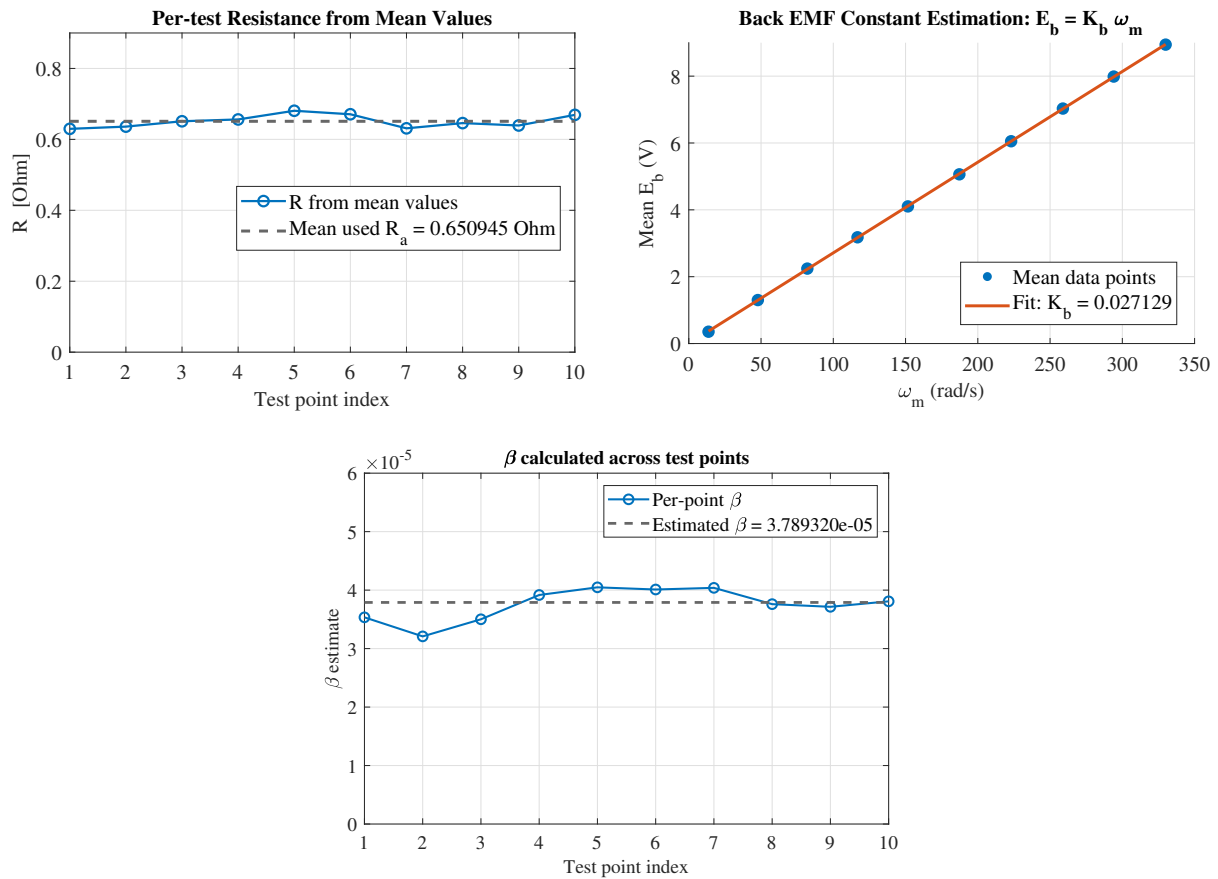


Figure 2: Plots obtained during the PMDC motor parameter estimation procedure.

Estimated generator torque T_{gen} came to 0.02558970 N.m

2. Estimation of Inductance

In this study, the armature inductance of the PMDC motor is estimated from the current ripple response obtained under bipolar PWM excitation. The motor is driven using an H-bridge operating in bipolar modulation mode at a fixed duty cycle. Refer Figure 3. During this test, the instantaneous motor terminal voltage and armature current are recorded and exported as a CSV file. The estimation method uses the current response during each constant-voltage switching interval of the PWM waveform.

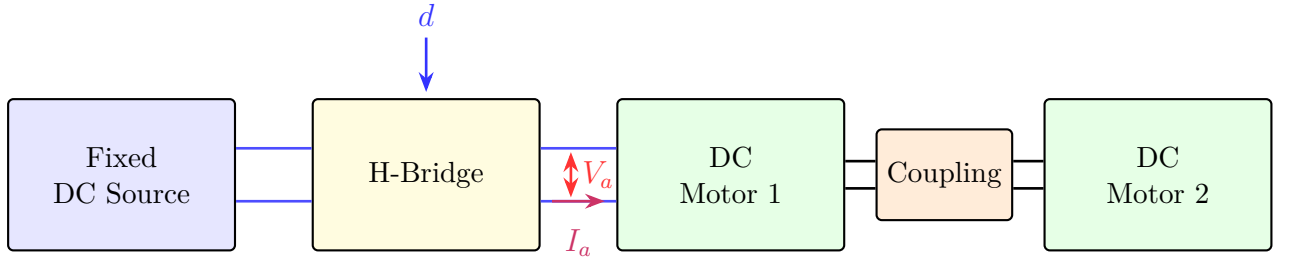


Figure 3: Block diagram of the inductance measurement setup

Unlike the steady-state open-loop test used for estimating R_a , K_b , and B , the present method makes use of the *dynamic current evolution* within each PWM sub-interval. Since the switching voltage repeatedly alternates between two levels, the current ripple within each constant-voltage interval follows the first-order RL dynamics of the armature circuit. By fitting the exact RL current response across multiple switching segments, the armature inductance L_a can be estimated.

2.1. Electrical Model Used for Estimation

The armature circuit of the PMDC motor is modeled as:

$$V_a(t) = R_a I_a(t) + L_a \frac{dI_a(t)}{dt} + E_b(t) \quad (18)$$

The measured voltage and current signals contain both:

- a slowly varying average component,
- a fast switching ripple component caused by PWM.

To isolate the ripple dynamics, the global mean of the measured voltage and current is computed as follows:

$$\bar{V}_a = \text{mean}(V_a(t)) \quad (19)$$

$$\bar{I}_a = \text{mean}(I_a(t)) \quad (20)$$

Then the ripple components are defined as:

$$v_r(t) = V_a(t) - \bar{V}_a \quad (21)$$

$$i_r(t) = I_a(t) - \bar{I}_a \quad (22)$$

This step removes the DC operating point and allows the estimation to focus only on the small-signal ripple behavior caused by PWM switching.

For a constant ripple voltage $V_{p,k}$ applied across an RL circuit with known resistance R_a , the current ripple satisfies:

$$R_a i_r(t) + L_a \frac{di_r(t)}{dt} = V_{p,k} \quad (23)$$

For the initial condition : $i_r(0) = i_{0,k}$

the exact solution is:

$$i_{r,k}(t) = \frac{V_{p,k}}{R_a} + \left(i_{0,k} - \frac{V_{p,k}}{R_a} \right) e^{-\frac{R_a}{L_a} t} \quad (24)$$

2.2. Global Multi-Segment Estimation of L_a

Suppose there are M valid constant-voltage segments. For the k -th segment:

- measured ripple current samples are $i_{r,k}(t_n)$,
- modeled ripple current samples are

$$\hat{i}_{r,k}(t_n; L_a) = \frac{V_{p,k}}{R_a} + \left(i_{0,k} - \frac{V_{p,k}}{R_a} \right) e^{-\frac{R_a}{L_a} t_n} \quad (25)$$

The estimation error for the k -th segment is:

$$e_k(t_n) = i_{r,k}(t_n) - \hat{i}_{r,k}(t_n; L_a) \quad (26)$$

The total cost function used in the script is the normalized sum of squared errors across all segments:

$$J(L_a) = \frac{1}{N_{\text{tot}}} \sum_{k=1}^M \sum_{n=1}^{N_k} \left(i_{r,k}(t_n) - \hat{i}_{r,k}(t_n; L_a) \right)^2 \quad (27)$$

where:

- M is the number of valid segments,
- N_k is the number of samples in the k -th segment,
- $N_{\text{tot}} = \sum_{k=1}^M N_k$ is the total number of samples used in fitting.

The armature inductance is then estimated as:

$$L_a^{(est)} = \arg \min_{L_a} J(L_a) \quad (28)$$

In the MATLAB implementation, this optimization is carried out numerically using a one-dimensional search based on `fminsearch`. Experimental results are shown in Figure 4

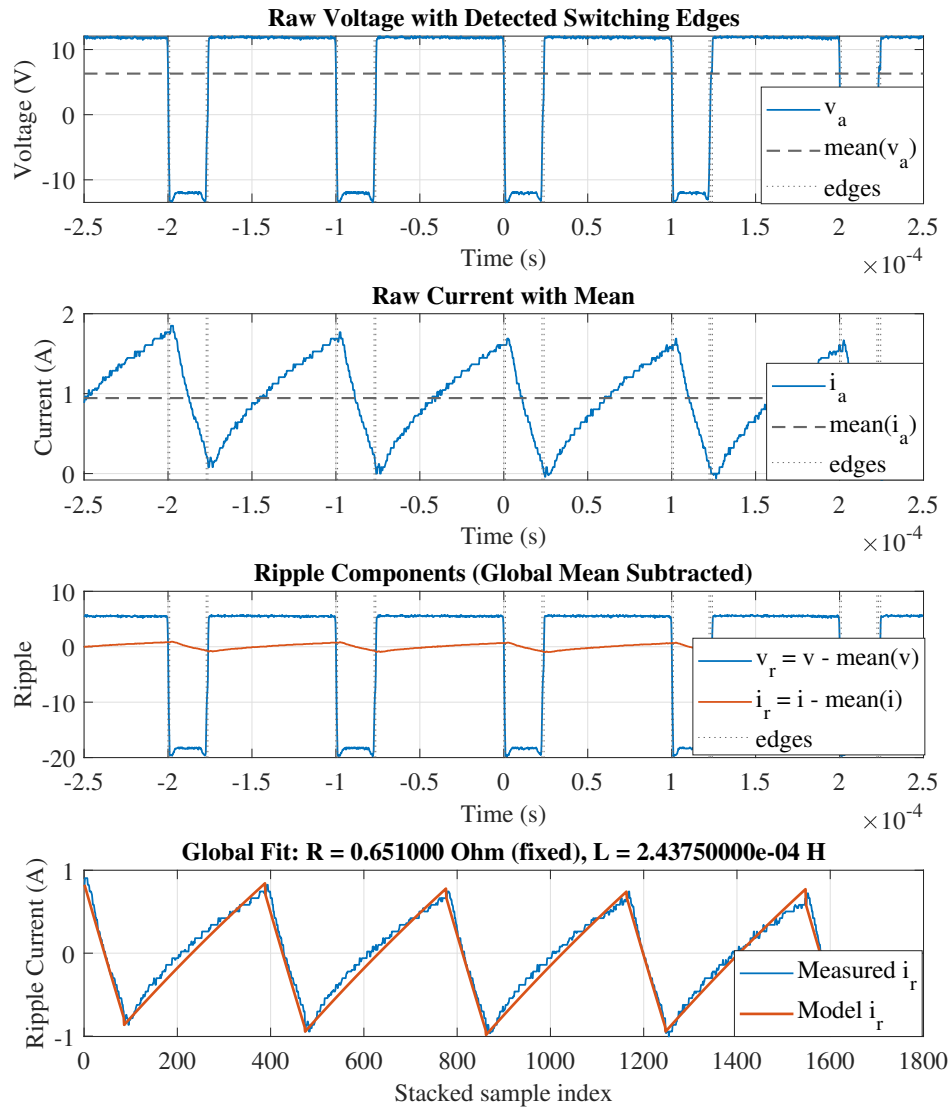


Figure 4: Inductance Estimation

3. Inertia Measurement

In this study, the total rotational inertia of the PMDC motor system is estimated from the open-circuit decay of the back-emf waveform after the motor supply is suddenly removed. The setup is arranged as in Figure 1. When the armature is disconnected and left open-circuited, the motor continues to rotate due to its stored kinetic energy, and the speed gradually decays because of viscous friction. Since the back emf of a PMDC motor is directly proportional to speed, the decay of back emf can be used to infer the mechanical time constant of the system and hence the inertia.

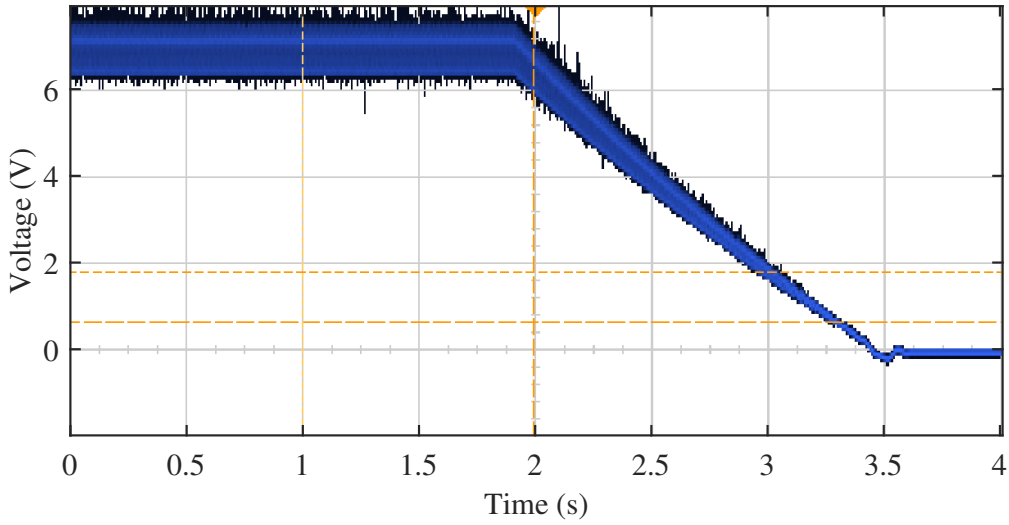


Figure 5: Experimental result : Decay of Back EMF, when input voltage is suddenly made to zero

In the present method, the motor is run near to its rated speed and the back-emf waveform is recorded after the input voltage is switched off. The back emf will gradually decay to zero, as shown in Figure 5, a Matlab script is written to compute the decay time constant from the experimental csv files. Once the decay time constant is obtained, the total rotational inertia is calculated using the previously estimated viscous friction coefficient B .

3.1. Mechanical Model After Switch-Off

Neglecting any constant external load torque, the post-switch-off mechanical dynamics are modeled as:

$$J \frac{d\omega_m}{dt} + B\omega_m = 0 \quad (29)$$

where:

- J is the total rotational inertia,
- B is the viscous friction coefficient,
- ω_m is the mechanical angular speed.

we know

$$\omega_m(t) = \omega_{m0}e^{-t/\tau_m} \quad (30)$$

where the mechanical time constant is:

$$\tau_m = \frac{J}{B} \quad (31)$$

Thus, if τ_m is known, the inertia can be computed as:

$$J = B\tau_m \quad (32)$$

This study was done for 5 speed values and the average value of " J " was used for further analysis

4. Conclusion

Table 2: Final estimated PMDC motor parameters obtained from experimental tests

Sl. No.	Parameter	Symbol	Estimated Value
1	Armature resistance	R_a	0.651 Ω
2	Back-emf constant	K_b	0.027129 V/(rad/s)
3	Viscous friction coefficient	B_m	3.789×10^{-5} N m s/rad
4	Generator/load torque	T_{gen}	0.02558970 N m
5	Armature inductance	L_a	243.75 μH
6	Armature electrical time constant	$\tau_a = \frac{L_a}{R_a}$	0.3744 ms
7	Total rotational inertia	J	2.461×10^{-4} kg m ²
8	Mechanical time constant	$\tau_m = \frac{J}{B_m}$	6.494 s